

Coaxial Helicopter design for Emergency Medical Services

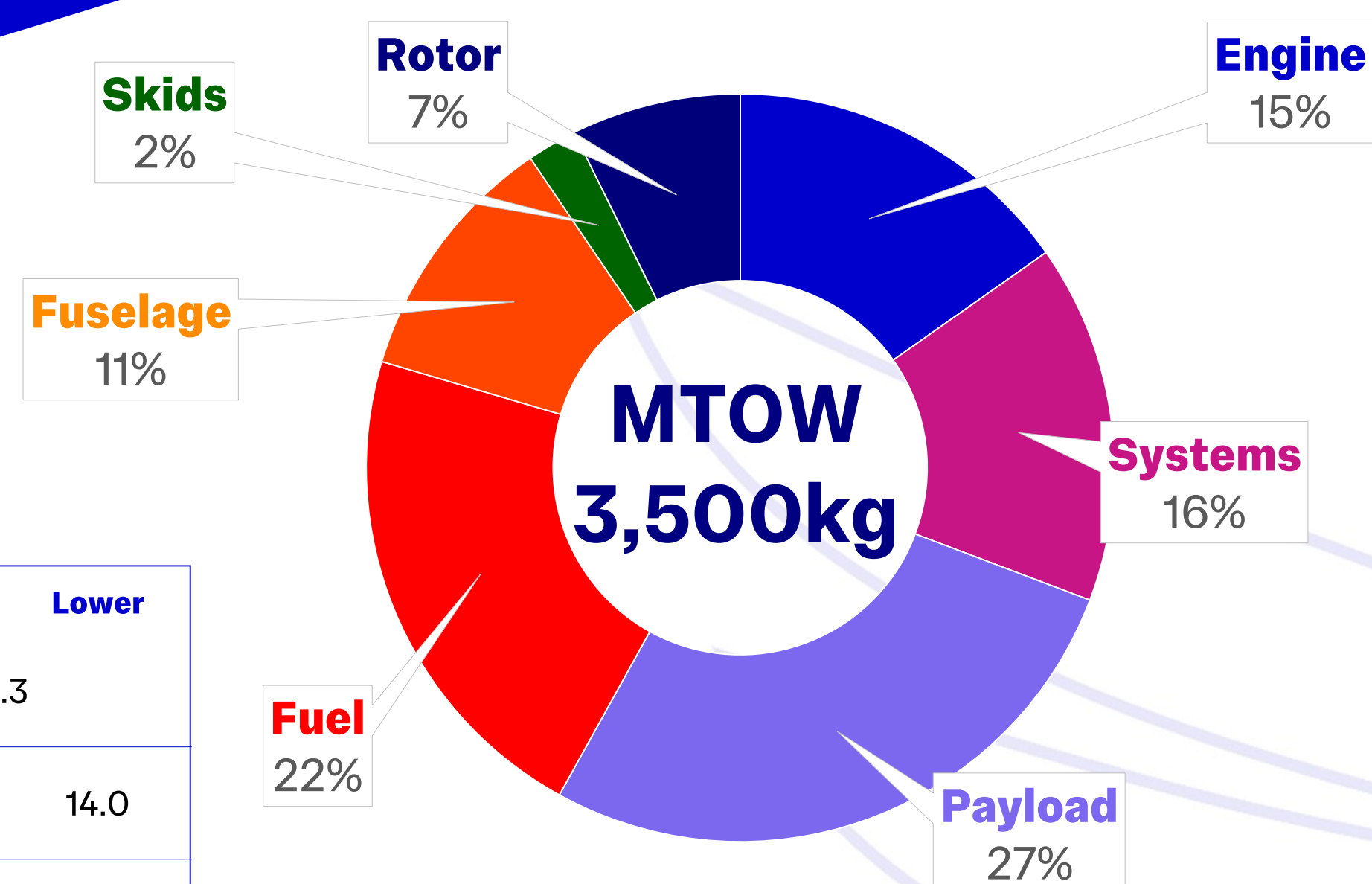
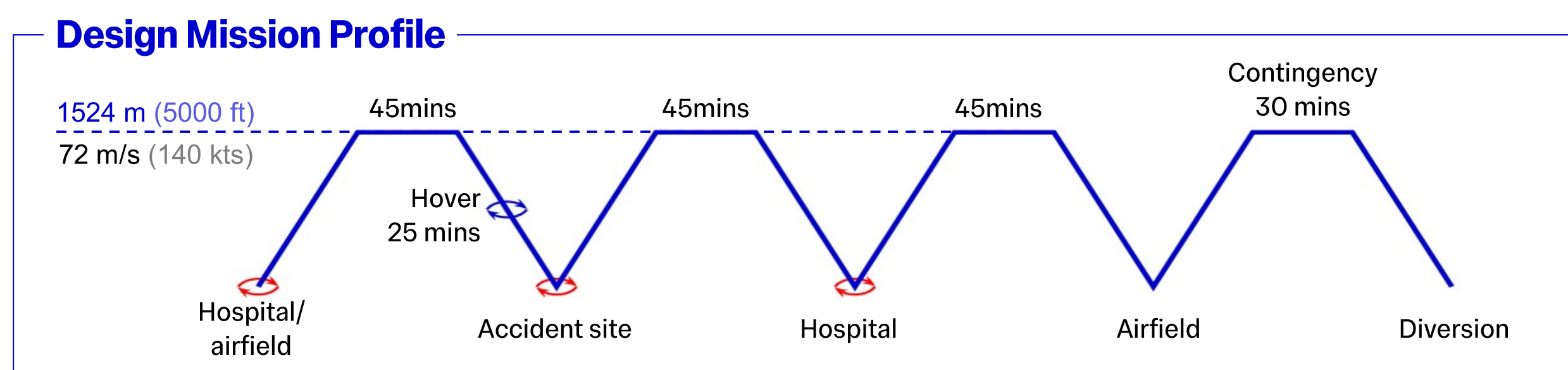
Engine	
Type	HTS900
Max power (kW)	1334
SFC (g/kWh)	319.45
Maximum torque (Nm)	2016
Fuel flow rate (kg/hr)	153
Transmission loss	7.8%
Transmission stages	3
Reduction ratio	15.82

Cabin	
MTOW (kg)	3500
Cabin volume (m³)	5.72
Fuel mass (kg)	700
Max pax	6
Skin material	QI CFRP
Skin thickness (mm)	2
Frame material	Al 7075
Heavy frames	2

Rotors		Upper	Lower
Rotor radius (m)		5.3	
Aspect ratio		20.0	14.0
Root chord (m)		0.265	0.416
Root aerofoil		RC(4)-10	
Tip chord (m)		0.265	
Tip aerofoil		RC(6)-08	
Twist (°)		-11	-6
N _B		3	
Root cutout		20%	
Operating RPM		396	
Blade material		UD CFRP + polyetherimide	
Flex beam material		UD CFRP	
Torque tube		17-4PH Steel	

Empennage	
Boom length (m)	0.2
Aerofoil	NACA 0015
chord length (m)	0.6
Thickness (m)	0.09
HTP width (m)	3
VTP width (m)	2
Empennage material	UD CFRP + Al 7075

Optimum Performance					
	Ceiling (m)	Hover Ceiling OGE (m)	Range (km)	Endurance (hrs)	Max speed (m/s)
No payload	7032	4232	1149	2.9	87.3
Max payload	6300	3701	1018	3.2	86.4



Structural Design

Through the use of **composites**, the weight of HALO is minimised.

A **semi-monocoque** layout was used, **with metallic alloy frames**/longerons for simplified manufacturing and CFRP skin.

CFRP was used for the rotor blades and tail as well. Aluminium alloy skids were selected over wheels for landing gear, due to their flexibility in landing on all types of terrain.

Aerodynamic Design

The first civilian coaxial design since the 1990s, **HALO bring new technology to old ideas**.

With **2 hingeless 3-bladed rotors** providing lift, **HALO** minimises rotor size and does away with the need for a tail rotor. Torque balance analysis was used to optimise blade twist and taper, maximising thrust and lift whilst minimising profile drag.

An **active flow control system** is also implemented to reduce drag by up to 10%. The fuselage geometry was optimised through iterative modelling using CFD whilst a H-tail layout minimises airframe size and allows control at high speed.

Engine Selection

The **Honeywell HTS900** was selected for HALO, delivering excellent performance with an excess of power for auxiliary systems.

A custom transmission minimises losses.

A helicopter designed from the ground up to save lives.

On average, **HEMS attends 126 patients a day across the UK.**

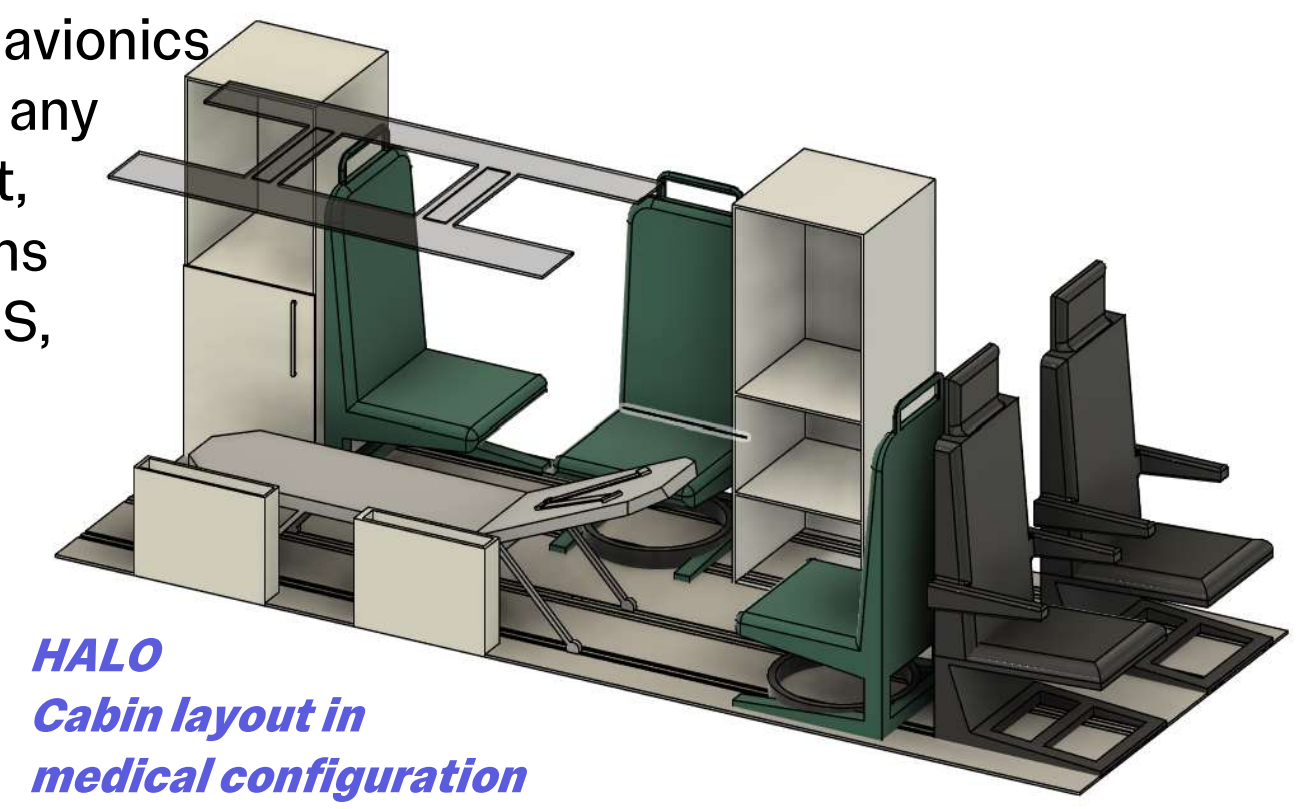
In an emergency, every second counts. That's why **HALO** is design for best-in-class performance in the light twin category. Certified under **CS-29**, our design aims to minimise time to the patient with an innovative design.

Flight and Medical systems

Onboard systems are designed for **maximum safety** whilst minimising mass and delivering value for customers. Redundancy is present in all flight and patient critical subsystems.

Modular, fully digital avionics enable safer flight in any weather, day or night, incorporating systems such as HTAWS, NVIS, NavFLIR, HUMS and ultrasonic deicing.

Cabin layout was done with regards to existing EMS vehicles, with provision for critical care equipment such as ECMO capability, prehospital analgesia, REBOA and other lifesaving procedures for the most urgent patients. Flexibility was maintained for other missions such as VIP transport or policing operations.



Control System Design

A **linearized state-space model** was generated following **trim analysis** of **HALO** with a plug-in physical model, enabling plug-and-play submodel updates without altering core functions.

The helicopter's static and dynamic stability was then evaluated to verify compliance with **CS-29** certification requirements

A **cascade-structured, PID-based SAS and autopilot** were developed: the altitude and airspeed loops, together with yaw- and roll-rate dampers, were **gain-scheduled** and tuned via a simplified mission profile to meet **ADS-33E-PRF Level-1** handling criteria

